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Taste-Elicited Activity in Facial Muscle Regions in 5–8-Week-Old Infants

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Abstract

The state of development of the sense of taste in humans during the first few months of life is only partially understood. Since taste plays a critical role in the feeding and nutrition of infants a better understanding of taste development during early life is required. Currently, information about the sense of taste in pre-verbal infants is obtained by analysis of videotaped facial expressions using the Baby FACS coding system. A potentially more objective faster procedure for assessing facial expressions not investigated in infants is electromyography (EMG). The method has been successfully used to study taste-elicited responses in the mid-face muscle regions of the levator labii and zygomaticus major of 6–9-year-olds and in a range of facial muscle regions in adults. Accordingly, this study aimed to investigate taste in young infants using EMG to 1) measure activity simultaneously in 4 facial muscle regions in response to 3 common tastants and 2) determine whether the activities of one or more muscle regions is needed to provide evidence of perception of a tastant by an infant. The results indicated that multiple facial muscle regions responded simultaneously but differentially to non-sweet and sweet tastants and recordings of activities from 3 or 4 regions simultaneously indicated that almost 100% of infants responded to the unpleasant tastes of quinine and citric acid, and 80% to sucrose.

Key words: electromyography, gustatory development, hedonics, infants, taste perception

Introduction

The state of development of the sense of taste in pre-verbal infants has been investigated using qualitative methods that have focused on reflex-like responses that include consummatory (Kajiura et al. 1992), sucking rates (Maone et al. 1990), and heart rate (Harper et al. 1982) responses. Such methods have provided evidence that at birth humans can perceive sweet, sour, and bitter tastants but not salt taste. However, at best these procedures only provide qualitative data on the level of taste development in an infant and are not suited for providing information, e.g., on taste sensitivity or for the existence or extent of a taste disorder an infant may be experiencing. More recently infant taste has been investigated using video-imaging to assess oro-facial responses in newborns. Early video studies suggested that oro-facial responses are innate and unlearned and that they are characteristic for sweet, sour, and bitter tastes (Steiner 1973). However, in a more recent study (Rosenstein and Oster 1988) it was reported that some of the proposed innate characteristic facial responses including "smiling" in response to sweet and "down-turned mouth corners" to bitter, were rarely observed. Unfortunately, obtaining reliable data using video-imaging requires substantial time for individual observers to be trained in the recognition of facial muscle movements, that is, in the recognition of Action Units (AUs) as in the Baby FACS coding system (Ekman and Friesen 1978), and for lengthy analyses of the videotapes. Importantly, the

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study by Rosenstein and Oster (1988) showed that none of the individual AUs encountered in the responses of newborns to tastes was highly taste-specific, even though some occurred more frequently than others, particularly in responses to the non-sweet tastes of bitter and sour. Furthermore, the latter study showed that certain facial actions and combinations of 2 or more facial actions in the brow/forehead and mid-face regions were common components of the responses to the non-sweet tastants including "brow raising" (AUs 1 and/or 2), "brow lowering" (AU 4), "cheek raising" (AU 6), "nose wrinkling" (AU 9), and "upper lip raising" (AU 10). These actions were all classified as representing non-specific negative responses. However, 2 mouth actions, namely, "lip puckering" (AU 18) and "mouth gaping" (AU 26 or 27) in response to sour and bitter, respectively, appeared to be more highly taste specific when they occurred with the negative brow and mid-face actions. The key point here is that each non-sweet tastant produced a pattern of responses that defined an infant's response. According to the authors, no single action unit could achieve this result.

Nevertheless, although the development of the Baby FACS coding system represented a major advance in studies of taste in infants, a more rapid and objective method that could potentially be used for analyzing facial expressions in infants is electromyography (EMG). It has been reported that EMG can sense facial muscle changes in response to a tastant that are not easily visible to an observer using video-imaging (Kring et al. 1999). In addition, the magnitude and duration of muscle activity can be quantitatively measured quickly as demonstrated in an EMG study with 6–9-year-olds (Armstrong et al. 2007). The study which involved the measurement of response thresholds of the zygomaticus major and levator labii muscle regions by measuring the magnitude of activity in the region of each muscle in response to 8 concentrations of 4 tastants, also identified the concentration of each tastant that induced the highest activity in each of these muscle regions. The latter study which is the only reported study of EMG responses to tastes in children, also showed that EMG can produce data that differentiates the responses of 2 muscle regions to sweet and non-sweet tastes. Thus, the findings that 1) combinations or patterns of muscle responses allowed identification of characteristic facial responses to sour and bitter tastes (Rosenstein and Oster 1988) and 2) the demonstration that monitoring 2 muscle regions using EMG produced data that differentiated between sweet and non-sweet tastes, prompted the design of the present study. In essence the design was based on using EMG to record simultaneously from a combination of muscle regions that embraced the 3 major regions of the face, namely, upper (brow), midface (nose, cheeks and lips) and lower (mouth and jaw), to capture the array of facial responses that are reported to be produced by sweet and non-sweet tastes. Accordingly, the 3 main aims of this study were 1) to demonstrate that a multi-electrode system can be used with 5–8-week-old infants to measure activity simultaneously in 4 facial muscle regions in response to each of the 3 common tastants, sucrose, citric acid, and quinine hydrochloride, 2) use this method to quantify the magnitude of the responses in each region and whether one or more muscles need to be monitored to determine the maximum number of infants who respond to each of the tastants, and 3) relate the arrays of electrical responses triggered by different tastants to specific facial behaviors and hedonic responses of infants.

Materials and methods

Participants

A total of 76 infants participated of which 43 were 6–10 weeks of age and participated in pilot studies and 33 were 5–8 weeks of age and were in the main study. All the infants were recruited from the Oxford and Paddington postnatal wards at The Royal Women's Hospital, Sydney within 2–3 days of birth. The relevant findings of the pilot study are described briefly in the section below on EMG.

The participants in the main study were 15 males and 18 females (mean age 6.5 weeks). Of these, 3 infants were too unsettled on the day of testing to participate. This age group was chosen since the Hospital and University Ethics committees would not approve studies with newborns until the method to be used had been proven with older infants. This is mentioned because a major long-term goal of the research programme with infants is to use EMG to assess taste function within 24 h of birth and during the non-verbal period of life. None of the participating infants had evident or reported nasal congestion or obvious oral or swallowing complaints. Written informed consent to participate in the study was obtained on a form from a parent or guardian. The study complied with the Declaration of Helsinki for Medical Research involving Human subjects. The research was approved by The University Human Research Ethics Committees (Approval No. HREC 04265) and the South Eastern Area and Illawarra Health Service Human Ethics Committee (Approval No. HREP 04/256).

Taste stimuli

The concentrations of taste stimuli used in this study were lower than in previous studies of taste and facial expressions in infants but were of moderate intensity when compared with those in psychophysical studies of intensity involving adults (McBride 1973; Wendin et al. 2011) or children (James et al. 2003; Laing et al. 2008; Laing et al. 2011). They were chosen for 2 reasons. First, in the only previous studies of taste-elicited facial expressions with infants (Steiner 1973; Ganchrow et al. 1983; Rosenstein and Oster 1988; Steiner et al. 2001), all of which involved videotaping of facial responses, the concentrations of tastants were mostly very to extremely strong. Indeed, to date the facial expressions reported for newborns in response to tastants have almost exclusively been obtained in response to strong stimuli and potentially may not be the expressions seen at more moderate levels. Secondly, the only EMG study that has provided information about taste concentrations that could act as a guide for studying taste in infants is the study by Armstrong *et al* (2007) who used 8 concentrations of both sucrose and quinine hydrochloride to determine the response thresholds of the levator labii superioris/alaeque nasi and zygomatic major muscle regions in 6–9-year-olds. The study also established the concentration of each tastant that produced the greatest activity in each muscle region. In the present study the concentration used for sucrose (0.205 M) (ex Sigma) was the one that produced the most common strongest response in the zygomaticus major region with 64.7% of children exhibiting their strongest response at this concentration (Armstrong et al. 2007). For quinine (0.0008 M) (ex Aldrich), the concentration was also selected from the latter study and was the concentration that produced a strong response in the levator labii region in the greatest number of children, that is, 47.1%. Since the latter study did not use citric acid as a stimulus, several sources of information involving taste intensity measures were consulted (McBride 1973; Laing et al. 2008; Wendin et al. 2011; Laing et al. 2011). From this information, a concentration of 0.01 M for citric acid (ex Analar) was chosen which provided a moderate intensity stimulus similar to the intensity of the other 2 tastants. Salt sensitivity was not assessed since initial salt perception is reported to emerge at approximately 4–6 months of life (Beauchamp et al. 1986). In addition, there has been no indication of a specific facial expression by newborns to salt (Rosenstein and Oster 1988), with newborns displaying indifferent responses to moderate concentrations of salt (Mistretta and Hill 1995).

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Purified drinking water (Nobles Ultra Pure Water, Sydney) was used as a diluent and the water control stimulus in the study. The tastants were refrigerated at 4°C overnight and taken out 2 h prior to tasting to equilibrate to room temperature (~ 22°C). All tastants were used within 48 h of preparation. Approximately 0.5 mL of the tastant (Ganchrow et al. 1983) was inserted between the lips and dropped onto the anterior third of the dorsal surface of an infant's tongue using a disposable 3 mL plastic bulb pipette.

Electromyography

The pilot study mentioned above in the section on Participants was needed to determine whether it was practical to locate electrodes in 4 muscle regions on the very small faces of the infants and whether facial responses could be recorded from the 4 regions. To record activity in a muscle region, the initial plan was to follow the convention of placing a pair of electrodes over each muscle region (Horio 2003) and another electrode behind the ear as an earth, with the difference in activity between the electrodes on a muscle defining the activity of a target muscle. However, when this methodology was trialed with 6–10-week-old infants, the small size of the face of each infant did not allow the placement of 8 electrodes on the 4 muscle regions chosen. However, given the importance of obtaining information from several muscle regions simultaneously to establish combinations of responding muscle regions, a different approach was investigated. Accordingly, on the advice of a neurologist, we measured activity in each of the 4 muscle regions by sitting one of each pair of electrodes at a location high on the brow which exhibited little or no activity in response to the tastants at the concentrations chosen for the main study. The other electrode remained on the muscle region of interest. Thus, a pilot study was used to indicate the brow location of the "silent" position' for the second electrode of each of the 4 pairs of electrodes. The data showed that the tastants produced little or no activity in each of the 4 regions of the so-called "silent muscle." For example, the mean responses to water (26 mV), sucrose (35 mV), and quinine (33 mV), were substantially below the mean responses of the levator labii muscle to water, sucrose and quinine (Table 2). Similar results were obtained with each of the other 3 muscle regions. Accordingly, 4 electrodes high on the brow were used, one for each of the other muscle regions studied (Figure 1).

The 4 facial muscle regions studied were those of the anguli oris, corrugator supercilii, levator labii, and zygomaticus major muscles. These regions were chosen to record activity in the upper face (brow/eyes), mid-face (cheeks, nose, lips, mouth), and lower face (jaws). Furthermore, the regions were chosen to include those reported by Rosenstein and Oster (1988) as regards the 2 key facial responses of "lip puckering" (AU18) and "mouth gape" (AUs 26 or 27) and other facial actions in the brow/forehead and mid-face regions as discussed in the Introduction. Thus the anguli oris region was chosen to measure activity in the lower face to detect the occurrence of negative responses including "mouth gape"(AU 26 or 27) and "downturn" of the mouth corners "lip corner depressor"(AU 15) in response to the bitter taste of quinine or sour taste of citric acid (Rosenstein and Oster 1988). The corrugator supercilii region recorded activity in the brow and eye regions which occur in response to the negative hedonic tastes of quinine and citric acid and include "brow lowerer" (AU4), "brow raising" (AU s 1 and 2),' knitted brows" (AU 4) (Tassinary et al. 1989) and various eyelid movements including "eyelid closing" (AU 7) and "eye squinching" (AU 44) (Tassinary et al. 1989). As regards the levator labii region unpleasant stimuli cause contraction of this muscle resulting in the negative expressions of "raising of the upper lip" (AU 10) and "wrinkling the nose" (AU 9) (Hu et al. 1999; Hu et al. 2000). Finally in the zygomaticus major muscle region the muscle inserts into the angle of the mouth and contraction of this muscle results in pulling lip corners up and back (Gray et al. 1974) as in the positive hedonic action of "smiling" AU (6 + 12).The zygomatic region also responds to "lip suck" (AU 28) in response to the sweet taste (Greimel et al. 2006) and "lip puckering" (AU 18) (Rosenstein and Oster 1988). Thus although a range of muscle regions were not specifically investigated here, each of the 4 muscle regions chosen have been shown to have overlapping activities with other nearby muscle regions. For example, Horio (2003) demonstrated using EMG that similar responses were obtained from the corrugator supercilii and orbicularis oculi regions in response to a wide range of tastants including the 3 tastants used in this study. Similarly, activities in the anguli oris region resembled those in the risorius and orbicularis oris regions. Tassinary et al. (1989) also investigated the responses of the corrugator supercilii region and found them to be similar to the nearby muscle region of the depressor supercilii/procerus, whilst the responses of the zygomatic major region were similar to the buccinators/risorius muscle regions. These findings indicate that although absolute identification of the muscle being sampled cannot be guaranteed at this stage of development of the method, recording from a muscle region together with responses in other regions of the face, potentially provides a qualitative measure of an infant's response to a tastant. Thus the power of the current

Figure 1. Positioning of electrodes to measure EMG activity in the 5 facial muscles used. The electrode on the lower jaw recorded activity in the anguli oris region (lower face), the electrode in the infant's lower left hand cheek recorded activity in the zygomaticus major region (mid-face), the electrode in the infant's right hand cheek recorded activity in the levator labii region (mid-face), the electrode above, and closest to the right eyebrow recorded activity in the infant's corrugator supercilii region (upper face). The 4 other electrodes located closest to the infant's hairline were the 'silent' electrodes. (Parental consent was given for the child's image to be published in the journal. A copy of the consent is held by the corresponding author).

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procedure described here is in its ability to record activities of combinations of muscle regions in response to tastants, measure the magnitude of a response in a muscle region, and produce data that can be used in multi-factorial statistics (Wendin et al. 2011) to provide an insight to the types of responses measured and to an infant's ability to perceive and differentiate tastants.

EMG of the activities of the facial muscle regions were recorded with a Mac Lab 8s and Bioamp ML 1321, with an MLA 1340 patient cable (AD Instruments, Sydney, Australia) attached to self-adhesive, disposable neonatal ECG electrodes (Kittycat, Tyco Health Care, Canada). The signals from the 4 electrodes from the facial muscle regions were amplified and filtered (high pass 10 Hz, low pass 5 kHz) and converted to waveforms at a frequency of 400 Hz. Channels 1, 2, 3, and 4 displayed the raw data obtained from the electrical activity recorded from the corrugator supercilii, levator labii, zygomaticus major, and anguli oris muscle regions, respectively. Channels 5, 6, 7, and 8 displayed the smoothed Root Mean Square (RMS) of the raw data which converted the EMG activity into a positive value. The 8 channels were needed to record and measure EMG activity of the 4 muscles regions simultaneously. For each RMSed EMG record the area under the curve for the 10-s following a stimulus or control (water) presentation was determined by integration.

Procedure

Subjects were tested either at the hospital or at their own home. Testing, including attaching the electrodes took anywhere from 30 min to 2 h depending on the behavioural state of the infant. All babies were fed before the test. Each infant received sweet, bitter and sour stimuli in 1 of the 6 possible pseudo random orders. The sequences of order of presentation of tastants and water which were repeated until all taste stimuli had been presented were water (rinse) → water (rinse) →10 s break → water (control) → 10 s break → sucrose or quinine or citric acid → water (rinse) etc.

This order ensured that there was a water control immediately preceding a tastant and each infant had a minimum 10-s interval between each control and taste stimulus to allow for recovery and a 20-s interval between a taste stimulus and the control to return to resting activity. The period of recovery and return to resting activity (20 s) was determined in pilot studies which indicated that baseline activity was achieved less than 10 s after stimulus presentation to the tongue of an infant. This recovery time agrees with the short duration of responses to tastants, for example, <5 s reported by Steiner et al. (2001). For example, the duration of individual facial responses including "nose wrinkle"—measured for the combination of the levator labii superioris and zygomaticus minor regions response occurred over 2 s, whilst "lip puckering" often lasted 3–5 s. These durations are all well within the recovery time employed in the present EMG procedure. In summary, EMG responses were recorded during 6 trials which consisted of one presentation each of sweet, sour, and bitter tastants and 3 presentations of water as the control. As regards repeating trials to determine the reliability of the method, in this study the narrow window of availability of the infants, the time taken for the placement and attachment of electrodes and the short time period the infants could be maintained in a restricted position before becoming too restless to record their response, precluded replicates being achieved at this stage of the research. To date, no study with young infants and newborns has obtained reliability information, for example, Steiner (1973), Ganchrow et al. (1983), Rosenstein and Oster (1988) and Steiner et al. (2001). However, by using over 30 infants compared to subject numbers of 8, 12, and 23 in the latter studies, this number was considered to be sufficient to achieve the goals of the study. In other words, the similarity of the data across infants was used to obtain an insight to the similarity of their responses.

Results

Activity of muscle regions

A summary of the EMG activity recorded in each muscle region in response to water and to each tastant is given in Table 1. To improve the precision of the values recorded for each tastant, the response to water in the trial preceding presentation of a tastant was subtracted from that of a tastant. Following this a 4 (muscles) × 3 (stimuli) within-subjects, ANOVA was used to determine if there were significant differences in EMG activity between the 4 muscle regions. Using an alpha level of 0.05, there was no main effect of muscle region F (3,87) = 1.93, P = 0.131, however, there was a significant main effect of tastant, F (2,58) = 8.15, P = 0.001, with sucrose generating significantly less activity compared to quinine (P < 0.001) and citric acid (P = 0.006) (Table 2). In addition, there were no tastant by muscle interactions, F(6,174) = 1.59, P = 0.152, indicating no particular muscle region responded with significantly greater

Table 1. Estimated marginal means and standard errors of EMG activities (Hz) for each muscle region in response to each tastant and water (n = 30)

Muscle region	Stimuli		SE	Muscle region	Stimuli		SE
Corr	S	695.54	136.19	Lev	S	2035.43	171.15
	WS	724.22	178.41		WS	1688.42	89.98
	Q	1168.67	213.71		Q	2388.33	223.69
	WQ	529.84	90.65		WQ	1910.20	177.73
	C	1201.71	286.23		C	2702.27	268.70
	WC	560.99	79.11		WC	2016.73	180.94
Zyg	S	1065.11	202.28	Ang	S	2195.55	231.04
	WS	969.43	179.36		WS	1855.94	223.77
	Q	1882.14	310.03		Q	3166.57	281.38
	WQ	669.09	132.87		WQ	1978.61	237.84
	C	1487.97	267.11		C	3211.62	304.57
	WC	865.95	171.46		WC	1921.10	253.19

Ang = anguli oris, C = citric acid, Corr = corrugator supercilii, Q = quinine hydrochloride, Lev = levator labii, Zyg = zygomaticus major, S = sucrose, SE = Standard error. The abbreviations for Corr, Lev, Zyg, and Ang are the same in Tables 1, 2, 4 and 5.

WS, WQ, and WC are the water controls for each stimulus.

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Table 2. Mean activity (Hz) in the facial muscle regions and percentage of infants responding to the taste stimuli

Muscle	Sucrose		Quinine		Citric acid	
	% Responders	Activity (mV)	% Responders	Activity (mV)	% Responders	Activity (mV)
Ang	50	556	73.3	1409	73.3	1425
Corr	26.7	190	70	660	46.7	746.9
Zyg	40	427	80	1223	70	841.6
Lev	40	400	56.7	645.5	56.7	749.4

activity than the others to any particular tastant. In brief, as shown in Table 1 the infants had substantially lower activities to sucrose which together with the finding that 6/30 infants showed no muscle activity to this tastant, suggests that a number of infants may not have perceived sucrose. Interestingly, 5/6 and 6/6 of the 6 infants who did not respond to sucrose gave strong responses to quinine and citric acid, respectively, indicating their sense of taste for bitter and sour tastes was functional. Indeed the magnitude of activities recorded for each of the 4 muscle regions indicated that the lowest activities were recorded when sucrose was the stimulus (Table 2). As regards the cause of the lower magnitudes of the responses to sucrose, concentration can be ruled out since all 3 tastants had similar moderate intensities. Other studies have also reported lower numbers of responses to sucrose compared to citric acid and quinine hydrochloride. For example, significantly lower numbers of facial responses were given to sucrose compared to the 2 non-sweet tastants as regards responses in the brow and mid-face (Rosenstein and Oster 1988). The lower number of individual facial responses could also have been due to the lower hedonic impact of sucrose compared to the unpleasant tastes of the non-sweet tastants. Indeed the magnitude of activities recorded for each of the 4 muscle regions indicated that the lowest activities were recorded when sucrose was the stimulus (Table 2). Given the relatively low power of the analysis (0.602), the data in Table 2 suggest that the activity of the zygomaticus major compared to that of the other muscles may indicate that it may be the only muscle differentiating between the 3 tastants. Finally, the zygomaticus major and anguli oris regions were found to have similar high activities in response to quinine compared to their responses to sucrose, indicating that they probably discriminated between these pleasant and unpleasant tastes.

Qualitative assessment of the number of infants responding to the tastants

A second analysis investigated the number of infants responding to a tastant. In this analysis, the criterion for defining when a muscle region had responded to a tastant was that the activity in the muscle region had to be larger than the activity recorded to water and larger than 100 mV. The latter activity value was chosen because pilot studies had shown that none of the tastants induced an activity of this magnitude in any of the "silent" muscles, the highest for an individual being 76 mV and the mean for the group being 31 mV. Using this criterion, the data shown in Table 2 indicates the percentage of infants responding in the 4 muscle regions to the tastants, whilst Table 3 indicates the individual infants whose facial muscle regions did or did not exhibit activity in response to the taste stimuli. In addition, Table 4 shows the number and frequencies of infants responding to 1, 2, 3, or 4 muscle regions to each taste stimulus. Taking all the data in the 3 Tables into consideration, the responses of the infants to each of the 3 tastants were assessed to determine whether they discriminated between the pleasant and unpleasant taste stimuli. The results are described below.

Infant responses to the unpleasant tastants quinine and citric acid

Quinine

Table 4 shows that measuring activity in the zygomaticus major, anguli oris, and corrugator supercilii muscle regions simultaneously (3 muscle combination), or measuring activity in all 4 muscle regions simultaneously, produces the greatest number of infant responders (96.7%) to quinine and this high number of responders provides strong evidence that infants can perceive quinine at 5–8 weeks of age. The results in Tables 3 and 4 are good indicators of the variability of the several facial muscles infants used in response to taste stimuli. For example, the zygomaticus major muscle region had the highest number of responders (80%) followed by the anguli oris (73.3%), corrugator supercilii (70%), and levator labii (56.7%) muscle regions (Table 2). The zygomaticus major and anguli oris muscle regions exhibited high EMG activity to quinine (Table 2) with activities of 1223 and 1409 mV, respectively, which suggests that the majority of infants were displaying strong visible actions in the mouth, cheeks and jaw which would likely have included the predominantly negative facial actions of "upper lip raiser" (AU 10), "cheek raiser"(AU 6) "sneer", and/or a "squinch-like expression"(AU 6) in the zygomaticus major region, and "gape" (AU 26 or 27) and/or "lip corner down" (AU 15) in the anguli oris region (Ekman and Friesen 1978; Rosenstein and Oster 1988; Steiner et al. 2001). Importantly, the high activity in the anguli oris region was expected since "mouth gape" is the classic response to the bitter taste of quinine (Rosenstein and Oster 1988) as is "lip corner down." The corrugator supercilii and levator labii muscle regions generated moderate activities of 660 and 645.5 mV, respectively (Table 2) and suggest that some infants were also displaying the negative facial actions of "brow lowering" (AU 4), "eye squinching" (AU 44), and/or "frowning" in the upper facial region and detected in the corrugator supercilii region and "nose wrinkling" (AU 9) and/or "upper lip raise" (AU 10) in the mouth region, respectively, and detected in the levator labii muscle region (Ekman and Friesen 1978; Rosenstein and Oster 1988; Steiner et al. 2001). Of the 6 infants who did not respond in the zygomaticus major region, 2 (infant 8 and 15), exhibited activities within the regions of the anguli oris, levator labii, and corrugator supercilii muscles, 2 within the anguli oris region (infant 16 and 24), 1 within the corrugator supercilii region (infant 14) and infant 17 failed to respond to quinine. Finally, a striking feature of Table 4 is that by monitoring activity in only 2 of the muscle regions reported to commonly respond to quinine, that is, anguli oris and corrugator supercilii, that 93.3% of infants were found to have responded to quinine. In contrast, only 63.3% of infants used these muscle regions to respond to sucrose.

Citric acid

As with quinine, the combination of activities in 4 muscle regions indicated that 29/30 infants (96.7%) responded to the sour taste of citric acid providing strong evidence that 5–8-week-old infants

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Table 3. Individual infants whose facial muscles responded or did not respond to the taste stimuli

Muscle	Lev			Zyg			Ang			Corr		
Infant	S	Q	C	S	Q	C	S	Q	C	S	Q	C
1	•	•		•	•	•	•	•	•			
2		•			•			•	•			
3	•	•	•	•	•	•		•	•	•	•	
4					•	•	•	•	•		•	•
5			•		•	•		•	•		•	
6			•	•	•		•	•	•			
7	•	•	•	•	•	•	•	•	•	•	•	•
8		•	•			•	•	•	•		•	•
9	•	•			•		•	•	•	•	•	•
10			•		•		•		•		•	
11		•			•	•		•	•		•	•
12					•	•	•					
13	•		•	•	•	•			•		•	•
14	•						•				•	
15	•	•						•		•	•	•
16	•		•			•	•	•				
17			•									
18					•		•	•	•		•	
19		•			•	•		•			•	
20	•	•	•	•	•	•	•	•		•	•	
21	•	•		•	•	•		•	•		•	•
22		•	•	•	•	•			•		•	•
23					•	•	•	•	•		•	•
24		•				•		•	•			
25	•		•	•	•	•		•		•		
26			•	•	•				•		•	•
27		•	•		•	•	•		•	•	•	•
28		•	•		•	•		•	•	•		•
29		•	•	•	•	•	•	•	•		•	
30	•	•	•	•	•	•		•	•		•	•
n	12/30	17/30	17/30	12/30	24/30	21/30	15/30	22/30	22/30	8/30	21/30	14/30
z	40%	56.7%	56.7%	40%	80%	70%	50%	73.3%	73.3%	26.7%	70%	46.7%

A dot indicates an infant responded. A blank indicates an infant did not respond.
n = number of individuals who responded. *z* = percentage of individuals that responded.

can perceive this stimulus (Table 4). Overall, the anguli oris (73.3%) and the zygomaticus major (70%) regions had the highest number of responders, followed by the levator labii (56.7%) and corrugator supercilii regions (46.7%) (Table 2) where only about half the infants responded. The anguli oris region exhibited similar high activity (1425 mV) to citric acid as it did to the unpleasant taste of quinine, suggesting that in this instance the prominent common facial expression of "lip pursing" (AU18) was sensed in the anguli oris region. This suggestion is supported by the report that the magnitudes of the EMG activity exhibited by the anguli oris and orbicularis oris regions in response to citric acid were very similar (Horio 2003). The high activity in the anguli oris region was predicted given that "lip pursing" was reported to be a common response of newborns (Rosenstein and Oster 1988). The zygomaticus major, levator labii, and corrugator supercilii muscle regions all exhibited moderate activities of 841.6, 749.4, and 746.9 mV, respectively (Table 2), suggesting that the majority of infants (70%) also displayed the negative facial actions "upper lip raiser" (AU 10), "cheek raiser" (AU 6), in the zygomaticus major region, brow lowering' (AU 4), 'eye squint (AU 44) and "frowning" in the corrugator supercilii region and approximately half may have been displaying "nose wrinkling" (AU 9), and/or "upper lip lift" (AU 10) in the levator labii region. Of the 8 infants who failed to show activity in the anguli oris region, 3 responded in the zygomaticus major and levator labii muscle regions (infants 16, 20, and 25), 2 with activities in the zygomaticus major region (infants 12 and 19), 1 with activity in the corrugator supercilii region (infant 15), 1 with activity in the levator labii region (infant 17,), whilst infant 14 did not respond to citric acid.

Sucrose

The significantly lower activities exhibited by the infants in response to the pleasant sweet taste of sucrose compared to the 2 unpleasant tastes of sour and bitter indicates that the discrimination, as suggested above, was largely based on the hedonic properties of the stimuli. For example, the low number of infants exhibiting responses in the corrugator supercilii region (26.7%) suggests facial actions in the brow and eyes were the least common in response to sucrose compared to the more common hedonically positive mouth actions that are likely to have involved activity in the zygomatic major and anguli oris regions (73.3%). From earlier studies these may have included the common response of "lip suck" (AU 28) involving the orbicularis oris muscle but detected here in the zygomaticus muscle region (Tassinary et al. 1989) and/or by the anguli oris muscle region (Horio, 2003). Also any parting of the lips or mouth opening that involved the orbicularis oris, for example, AU 8, could have been detected in the anguli oris region (Horio 2003). "Smiling"

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(AU 6 + 12), commonly exhibited by infants in response to sucrose (Steiner et al. 2001) could have generated activity in the zygomatic major region due to the act of drawing the lip corners up and back (AU 12), and "upper lip lift" (AU 10) in the levator labii muscle region which is known to contribute to a "smile" (Steiner et al. 2001). Most importantly, however, even though the magnitudes of the activities recorded in response to sucrose were significantly lower than for the 2 unpleasant tastants, the locations of the electrodes in the brow, mid-face and lower face, enabled a high number of infants (80%) to be identified as responding to sucrose.

In summary, taking both the individual response rates and mean EMG muscle region activities into consideration, the data suggests that a sensitive procedure for assessing taste function in infants is to simultaneously monitor the activity of the zygomaticus major, anguli oris and corrugator supercilli muscle regions or all 4 muscle regions simultaneously for bitter and sweet perception and all 4 muscles simultaneously for sour perception. This provides a very high response level to quinine and citric acid (96.7%), and a high response level (80%) to sucrose. In addition, the data suggest that the facial expressions likely to be associated with the various stimulated muscles were hedonically positive for sucrose and negative for quinine and citric acid.

Principal components analysis

A principal components analysis (PCA) was used to examine the association between the EMG activity in the 4 facial muscle regions of the anguli oris, zygomaticus major, corrugator supercilii, and levator labii and the 3 taste stimuli, across the 30 infants. The extraction of multiple Factors or alternatively the identification of common Factor loadings could then be interpreted as evidence for 1) similar behavioral responses, that is, similar muscle activities to the 3 tastants or 2) the existence of independent responses of muscle regions to the tastants. Accordingly, a PCA with varimax rotation was performed on the twelve facial muscle regions and tastant conditions. There was no missing data and no outliers. Evaluation of assumptions was generally satisfactory, although KMO was 0.419, which as expected from previous analyses suggests that the number of participants was low (Field, 2005). A Scree plot of the data indicated that a three-Factor solution was the best option, accounting for 53.72% of the variance. The component loadings, communalities (*h2*), and percentage of variance explained after varimax rotation are shown in Table 5.

As can be seen in Table 5, variables with a strong loading (>0.5) on component 1 appear to reflect a hedonic grouping of responses, with the 2 unpleasant tastants quinine and citric acid activating the corrugator muscle region clustered together. In contrast, the clustering of responses to the pleasant taste of sucrose activating the anguli oris and zygomaticus muscle regions appears to be at the opposite end of a hedonic axis as indicated by the negative Factor loadings. The latter "hedonic axis" is indicated in Figure 2.

Thus the 2 distinctly different taste sensations at the ends of the "hedonic" axis are likely to have generated 2 very different sets of facial expressions. First, since both of the corrugator region responses to bitter and citric acid would have resulted from unpleasant sensations, they are very likely to have been observed in the brow and eye regions with possible facial responses of "brow lowering" (AU4) and "eye squint" (AU 44) or "lid tightener" (AU7) (Ekman and Friesen 1978). The second set of facial expressions at the other end of the hedonic axis representing the pleasant taste sensation of sweet are likely to have been caused by responses in the mouth region due, for example, to the lip corners being pulled obliquely upwards by the zygomaticus major muscle towards the cheek bones giving the appearance of a "smile" (AU12) and in other infants "lip suck" (AU 28) may have been the positive hedonic response involving the orbital muscles pulling the lips into the mouth (Ekman and Friesen 1978) and sensed in the adjacent anguli oris region (Horio 2003). The finding that the PCA revealed the existence of a hedonic axis is a significant one since it strengthens the proposal that the infants discriminated the pleasant and unpleasant tastants on the basis of hedonics not concentration or quality features. Furthermore, in contrast to the ANOVA described above which did not find differences between the magnitudes of the muscle region responses, the PCA revealed that there were differences between the hedonic responses to the pleasant and unpleasant tastants.

Table 4. Frequencies and number of infants responding in 1, 2, 3, or 4 facial muscle regions to sucrose, quinine, and citric acid

Muscle combination		Tastant					
		Sucrose		Quinine		Citric acid	
		n	%	*n*	%	*n*	%
One muscle							
zyg		12/30	40	24/30	80	21/30	70
ang		15/30	50	22/30	73.3	22/30	73.3
corr		8/30	26.7	21/30	70	14/30	46.7
lev		12/30	40	17/30	56.7	17/30	56.7
Two muscles							
zyg	ang	22/30	73.3	28/30	93.3	27/30	90
	corr	16/30	53.3	27/30	90	25/30	83.3
	lev	16/30	53.3	27/30	90	25/30	83.3
ang	corr	19/30	63.3	28/30	93.3	23/30	76.7
	lev	21/30	70	24/30	80	26/30	86.7
corr	lev	14/30	46.7	25/30	83.3	22/30	73.3
Three muscles							
zyg ang corr		24/30	80	29/30	96.7	28/30	93.3
zyg ang lev		22/30	73.3	28/30	93.3	28/30	93.3
zyg corr lev		19/30	63.3	28/30	93.3	27/30	90
Four muscles							
zyg ang corr lev		24/30	80	29/30	96.7	29/30	96.7

n = Number of infants responding. % = percentage of infants responding.

Table 5. Varimax rotated component loadings for 3 factors for facial muscle regions and taste stimuli combinations

Muscle and stimuli condition		Factors			
		1	2	3	*h2*
corr	Citric	0.81			0.88
corr	Quinine	0.78			0.60
ang	Sucrose	−0.61			0.54
zyg	Sucrose	−0.49	0.43		0.44
zyg	Quinine		0.73		0.60
zyg	Citric		0.73		0.66
ang	Citric		0.66		0.48
ang	Quinine		0.60		0.47
lev	Citric			0.75	0.61
lev	Sucrose			0.66	0.50
lev	Quinine			0.63	0.44
corr	Sucrose			0.43	0.21
% of variance		18.98	18.73	16.01	53.72

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The variables loading strongly on component 2 (Table 5), which are represented by the black circles in Figure 3, appear to be indicative of strong unpleasant responses to bitter and sour in the zygomaticus major and anguli oris muscle regions. Therefore, Factor 2 could be labelled "muscle responses to unpleasant tastes." Since the corrugator supercilii region is not featured in Factor 2, the facial response is likely due to changes in the mouth and cheeks or jaw. The anguli oris region responded strongly to the bitter and sour tastes with high activities of 1409 and 1425 mV, respectively, with the same high frequency of 73.3% of infants responding to both (Table 2). As proposed above, it is likely that the response to bitter sensed in the anguli oris region was the characteristic "mouth gape" (AU 26 or 27), whilst this region responded to the facial response "lip corner down" (AU15) and "lip pursing" (AU 18) in response to the sour taste. Similarly, the zygomaticus major region exhibited high activities to bitter and sour tastes, recording respective values of 1223 and 841.6 mV and response frequencies of 80 and 70%. Since the latter muscle region is primarily involved with facial responses in the mouth and cheeks it may have sensed the negative hedonic response of "lip pursing" (AU 18) (Ekman and Friesen 1978), a common response to the sour taste. As regards the response to the bitter taste the mid-face action unit "cheek raiser" (AU 6)(Rosenstein and Oster 1988) for which the zygomaticus major region is involved, is a likely negative facial response. The high frequency of response of the combination of the zygomaticus muscle region and anguli oris region for bitter and sour, that is, 93.3 and 90.0%, respectively, also supports the proposal that both the anguli oris and zygomaticus major regions in combination are largely responsible for the infants responses to these 2 unpleasant tastants and that like 6–9 year olds (high zygomatic major activity) (Armstrong et al. 2007) they found them to be unpleasant stimuli.

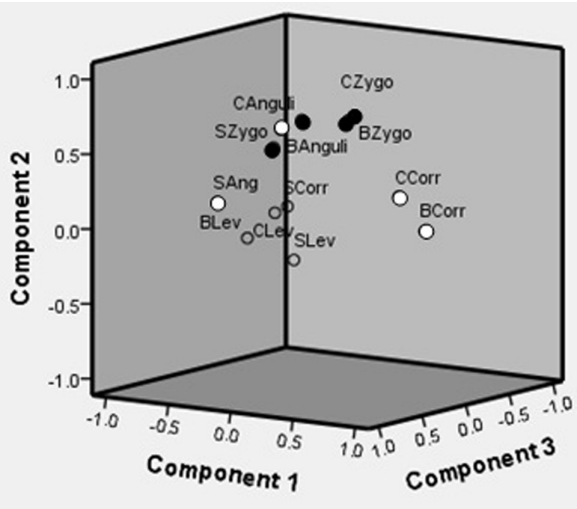

Figure 2. Factor loading plot showing the 3 Factors: hedonic Factor 1 (white circle), strong unpleasant responses Factor 2 (black circle), and levator Factor 3 (clear circle).

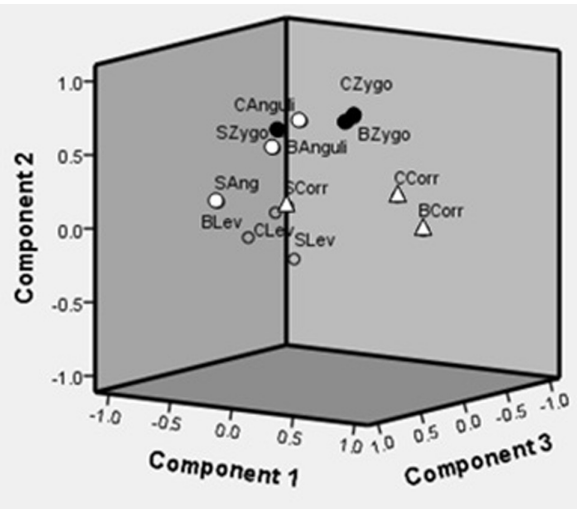

Figure 3. Factor loading plot showing the separate muscle clusters for bitter (B), sour (C) and sweet (S) stimuli in the anguli oris (white circle), zygomaticus (black circle), levator labii (clear circles), and corrugator muscles (triangles).

Why the response of the zygomaticus major region to sweet is listed as part of Factor 2 is difficult to interpret. In the Factor loading plot (Figure 2), the zygomaticus major region responses to sweet are located adjacent and closer to the very unpleasant region of bitter and sour, than to the sucrose responses of the anguli oris, levator labii and corrugator supercilii muscle regions. However, at 5–8 weeks of age it is possible that the response to the pleasantness of sweet is part of the facial actions of a "smile" (AU 12) and part "sneer" due to incomplete development of the 2 facial expressions "lip corner raiser" (AU 10) and "upper lip raiser" (AU 12) and differentiation of the 2 may be difficult. Alternatively, Rosenstein and Oster (1988) reported in video-imaging studies that newborns displayed a negative facial action to the sweet taste of sucrose, which later turned into relaxation and this may be what occurred here.

The variables loading on component 3 appear to be comprised of the muscle regions that had low numbers of responding infants and low activities compared to component 2. Given that the contribution of the corrugator supercilii region is very low with only 26.7% of infants showing responses and an activity of 190 mV which was not far above the response criterion of 100 mV, it can be concluded that the main contributor to this factor is the weak responses to all 3 tastants in the levator region. For simplicity, the factor could be labelled the "levator factor." Interestingly, a common feature of the facial actions in both muscle regions is that stimulation can deepen the nasolabial furrow (Ekman and Friesen 1978). Furthermore, 6/10 of the infants who responded with activity in the corrugator supercilii region also showed activity in the levator labii region indicating it is quite possible that a "frown" was part of the response of the infants. Importantly, Figure 3 shows the 4 different clusters of the levator labii (clear circles), corrugator supercilii (triangles), zygomaticus major (black circles), and anguli oris (white circles) muscle regions, indicating that each region has its own distinctive facial responses. However, the fact that each Factor involves at least 2 muscle regions (Table 5) indicates that there is an overlap of muscle region activity and that infants are using more than one facial muscle action in response to pleasant and unpleasant taste stimuli. An overlap of facial muscle actions was also found in video-imaging studies in newborns to taste stimuli (Ganchrow et al. 1983; Rosenstein and Oster 1988).

Discussion

The 3 main aims of this study, namely, to demonstrate that a multi-electrode system can be used to measure activity simultaneously in 4 facial muscle regions in response to each of 3 common tastants, determine whether one or more muscle regions need to be monitored to identify the maximum number of infants who respond to 3

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common tastants, and relate the arrays of electrical responses triggered by different tastants to specific facial behaviors and hedonic responses of infants, were achieved. In particular, the study showed that 5–8-week-old infants use a variety of facial muscle actions whose activities can be sensed and recorded by siting electrodes over the anguli oris, corrugator supercilii, levator labii, and zygomaticus major muscles in response to tastants and reflect whether a tastant is pleasant or unpleasant. Secondly, simultaneous EMG recordings of the activities in the regions of the 4 muscles showed that almost 100% of infants responded to the unpleasant taste of quinine and citric acid, and to a lesser extent the pleasant taste of sucrose (80%). Thirdly, PCA indicated that 3 main factors underlie the facial responses of infants to the 3 tastants, namely a hedonic factor, an unpleasant factor and a levator labii-related factor.

Thus, by recording from several muscle regions simultaneously the study not only achieved the important aim of demonstrating that this was possible with 5–8-week-old infants, but it also identified muscle regions infants use to respond to pleasant and unpleasant tastants. In doing so, it provided an insight to the likely facial expressions involved in response to particular tastants. In essence, the composition of the muscle region arrays in which activity was triggered by the individual tastants and the magnitude of the activity in the individual muscle regions, provided a unique insight to the behaviour elicited by each tastant. Thus infants produced high activities particularly in the zygomatic major and anguli oris muscle regions to the unpleasant tastes of quinine and citric acid, and much lower activities in the 4 muscle regions to the pleasant taste of sucrose (Table 2). The latter result is similar to that reported by Horio (2003) in a study with adults who found that citric acid and quinine hydrochloride produced higher EMG activity than sucrose in 8 muscle regions across the face including the corrugator supercilii and the anguli oris regions. Also Armstrong et al. (2007), in a 4-replicate EMG study, found lower activity in the levator labii muscle region in response to sucrose compared to quinine hydrochloride, citric acid and sodium chloride in 6–9-year-olds. The latter data, therefore, provide strong support for the appropriateness of the locations of the electrodes to record infant's responses to the tastes described in earlier studies with newborns. In addition, the zygomaticus major and anguli oris muscle regions exhibited high EMG activity to quinine (Table 2) with activities of 1223 and 1409 mV, respectively. The latter result suggests that the majority of infants were displaying strong visible actions in the mouth, cheeks, and jaw that most likely included the predominantly negative facial actions of "upper lip raiser" (AU 10), "cheek raiser" (AU 6) "sneer" and/or a "squinch-like expression" (AU 6) in the zygomaticus major region, and "mouth gape" (AU 26 or 27) and/or "lip corner down" (AU 15) in the anguli oris region (Ekman and Friesen 1978; Rosenstein and Oster 1988). Importantly, the high activity in the anguli oris region was expected since "mouth gape" is a common response to the bitter taste of quinine (Rosenstein and Oster 1988; Steiner et al. 2001). Indeed, Rosenstein and Oster (1988) referred to mouth gape as "taste specific" when it occurred in the context of negative brow and midface actions including some mentioned above. The corrugator supercilii and levator labii muscle regions generated moderate activities of 660 and 645.5 mV, respectively, to quinine (Table 2) which suggests that some infants were also displaying the negative facial actions of "brow lowering" (AU 4), "eye squinching" (AU 44), and/or "frowning" in the upper facial region detected in the corrugator supercilii region, and "nose wrinkling" (AU 9) and/or "upper lip raise" (AU 10) in the mouth region, respectively, and detected in the levator labii muscle region (Ekman and Friesen 1978; Rosenstein and Oster 1988; Steiner et al. 2001). The high activity in the anguli oris region (1425 mV) and a high number of responders (73.3%) in response to citric acid was also expected since "lip pursing" (AU 18), is a commonly seen "taste specific" response (Rosenstein and Oster 1988) to citric acid when there are certain brow and midface actions occurring during the response period. Thus, although "lip pursing" originates from the orbital muscles of the mouth, for example, orbicularis oris (Ekman and Friesen 1978) it has been demonstrated that the magnitudes of the EMG activity exhibited by the nearby anguli oris and orbicularis oris in response to citric acid are very similar (Horio 2003). The zygomaticus major, levator labii, and corrugator supercilii muscle regions exhibited moderate activities of 841.6, 749.4, and 746.9 mV, respectively (Table 2), to citric acid, suggesting that the majority of infants (70%) also displayed the midface negative facial actions "upper lip raiser" (AU 10) and/or cheek raiser' (AU 6), in the zygomaticus major region, and the upper face actions of brow lowering (AU 4) and or 'eye squint (AU 44) in the corrugator supercilii region and approximately half (56.7%, Table 2) may have been displaying "nose wrinkling" (AU 9), and/or "upper lip lift" (AU 10) in the levator labii region. Importantly, the muscle actions described above, used in response to the 2 categories of tastants are in accord with those reported by others (Rosenstein and Oster 1988; Steiner et al. 2001) using the video-imaging technique.

As regards the combinations of facial muscle regions responding to sucrose, this was more difficult to predict from previous reports of the responses of newborns obtained by video-imaging and FACS. Different authors have reported different facial reactions. The question of whether infants smiled or showed actions resembling a "smile" in response to sucrose was particularly contentious (Rosenstein and Oster 1988). However, because a number of facial actions have been described and have reasonable expectations of accompanying sucrose tasting, we predicted the following. First, a number of authors describe "lip suck" (AU 28) (Ganchrow et al. 1983) which involves the orbital muscles surrounding the mouth and lips (Ekman and Friesen 1978). As described above, activity in the muscles of the mouth and lips involving the orbicularis oris has been shown to be also sensed by the anguli oris, As shown in Table 2, 50% of infants exhibited moderate activity in the anguli oris region in response to sucrose. Secondly, the facial actions of "lifting up of lip corners" or "smile" (AU 12) (Ekman and Friesen 1978), are reported to be produced by the action of the zygomatic major muscle pulling the lip corners obliquely upwards towards the cheekbones (Ekman and Friesen 1978). In this study, 40% of infants displayed moderate activity (427 mV) in the zygomaticus region in response to sucrose. Since the levator labii muscle is also attached to the upper lip, and it has been attributed to contribute to the facial action "sneer" (Ekman and Friesen 1978) it can also be expected to respond to either of the 2 facial actions described above. In brief, the locations of the electrodes based on the findings from FACS facial expressions (Ekman and Oster 1978; Rosenstein and Oster 1988) and gustofacial response studies (Steiner 1973; Ganchrow et al. 1983) appeared to provide a unique qualitative insight to the responses to tastes by infants.

The evidence presented above as regards the activity in muscle regions associated with negative hedonic responses when quinine or citric acid were the stimuli and positive hedonic responses when sucrose was the stimulus, together with the discovery of a "hedonic axis" as a major factor in the factor analysis, indicates that the infants discriminated between the pleasant and unpleasant tastants on the basis of their hedonic properties. Using PCA as occurred here, Wendin et al. (2011) also found a spatial separation of facial

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expressions attributed to sucrose and the 2 unpleasant tastants caffeine (bitter) and citric acid, indicating that the hedonic properties of tastants feature in the discrimination of pleasant and unpleasant tastants. These are important findings since they indicate that factor analysis can provide an insight to the relationship between a tastant, the activity generated in several muscle regions and whether the facial action(s) associated with these regions represent positive or negative hedonic reactions. Currently, as demonstrated here with a 4-electrode EMG system, high numbers of infants were identified as having perceived the 3 tastants using muscle regions that would be expected for their hedonic characteristics. Clearly, more information is required to identify the locations on the infant face that produce optimum response characteristics, particularly as regards the amount of activity and duration of activity to maximize the level of responses found. Studies of the range of activities of a muscle have been reported with adults (Tassinary et al. 1989; Horio 2003) and have provided an insight to the size of an area of the face exhibiting activity from a particular muscle region and the identity of other muscles that also sense the activity. Such information would be invaluable for pinpointing the most relevant muscles here.

The present finding of a high number of infants (96.7%) perceiving quinine and citric acid is similar to the high response rate found for 6–9-year-old children (97.1%) who responded to the latter tastants with increased activity in the zygomatic major muscle region (Armstrong et al. 2007). However, comparison of the activities obtained in the 2 studies, 228.3 versus 1223 mV for the older and younger children, respectively, is not possible for 2 reasons. First, in the earlier study activity was recorded for only 5 s after a tastant had been presented compared to 10 s in this study. Secondly, the 2 recording electrodes were placed bilaterally on the face of the older children compared to a single electrode on the recorded muscle and another high on the brow in this study. Since the magnitude of activities obtained using the 2 recording procedures for measuring responses to tastants has not been reported, it is not possible to use the activity difference for the zygomaticus major muscle region as an indicator of different responses for the 2 age groups. As indicated in the Methods, it was necessary to modify the recording method here because it was not possible to fit bilaterally positioned electrodes on the smaller faces of the infants. However, the finding here that activity in the zygomaticus major region (1223 mV) was higher than in the levator labii region (645.5 mV) mimicked the earlier result where activity in the zygomaticus major region (228.3 mV) was greater than for the levator labii region (139.4 mV), suggesting that the facial expressions in response to quinine may have been similar.

As regards the lower number of infants (80%) in this study perceiving sucrose compared to the other 2 tastants and to sucrose (100%) in the study with 6–9-year-olds (Armstrong et al. 2007), this may have been partly due to the concentration of sucrose used here being lower than ideal. For example, in the earlier study the concentration used with each child was the optimum response concentration for that child, whereas here the concentration was the average optimum response concentration (0.205 M) established with the older children. Due to the limited availability of the infants and the duration required to establish their optimum response concentration, it was not possible to obtain this information in the present study. An alternative explanation could be that the pleasant taste of sucrose simply does not stimulate as much activity in facial muscles as the 2 unpleasant tastants.

The concentration used here with infants was much lower than those used in the main studies with newborns 0.3 M (Steiner et al. 2001), 0.73 M (Steiner 1973; Ganchrow et al. 1983; Rosenstein and Oster 1988). However, in none of these studies was it proven that the concentration was the optimum one. Because these are very high sucrose concentrations caution was taken here not to use too high a concentration since it has been reported that the viscosity of sucrose increases above 0.5M and introduces doubt as to whether a subject is responding to the sweet taste or the particular viscosity. Another potential complicating factor as regards concentration choice is that development of sweet taste perception may be occurring over the first year of life. For example, Desor et al. (1977) reported that sweet acceptance in newborns and infants remains stable from the ages of 1–3 days and 20–28 weeks, whilst according to Beauchamp et al. (1986) acceptance decreases between the ages of 4 and 6.7 months. Another report indicates there are small decreases in sweet acceptance between the ages of 6 and 12 months, shifting from preference to indifference when compared to water (Schwartz et al. 2009). The latter authors also suggested that acceptance may differ between individual infants, suggesting developmental changes in sweet acceptance or preference occur in the first year of life and that acceptance or preference for sweet does not seem to be as stereotyped as it has been represented in young infant studies of facial expressions. Other indirect evidence that taste function may be undergoing development in 5–8-week-olds are the changes that occur in salt perception. For example, initial salt perception emerges at 4–6 months of age and develops to adult level by approximately 2 years of age (Beauchamp et al. 1986). Finally, the finding that 20% of the infants did not respond to sucrose may have more to do with the reliability of the test than any other cause. Currently, the reliability of the results has not been established. However, given that very high numbers of infants responded to the other 2 tastants, that is, 29/30, this reason seems less likely compared to the cause being due to insufficient development.

The important aim of determining whether infants discriminate pleasant and unpleasant tastants was achieved even though the ANOVA results showed no particular muscle region responded to one tastant more than any other. Nevertheless, the activity data showed that the magnitude of the activity in all 4 muscle regions to sucrose was significantly lower than to quinine and citric acid. The latter finding was further supported by information about the responses of the infants from the analysis of the EMG data using PCA, which indicated that there were 3 factors underlying the responses. These were primarily related to the hedonic nature of the stimuli and the type of muscle regions involved in the responses. Specifically, one was a hedonic factor reflected mainly by activity in the corrugator muscle region in response to the unpleasantness of quinine and citric acid, and in the levator and zygomaticus muscle regions to the pleasantness of sucrose. The latter factor indicated that the infants had discriminated between the pleasant and unpleasant tasting stimuli. The second factor was indicative of strong unpleasant responses to bitter and sour tastes by the zygomaticus and anguli oris muscle regions. Such responses did not occur in response to sucrose. A third factor was mainly comprised of the responses of the levator labii region to the 3 tastants. However, since the magnitude of the activities of the latter muscle region to the 3 tastants were similar and low (Table 2) and fewer infants used this region to respond to the tastants, it is unlikely the expression(s) produced in the levator labii muscle region were as marked or contributed as greatly to the overall facial responses of the infants as the other 3 muscle regions. The responses of the levator labii muscle region, therefore, contributed least to the discrimination of the tastants. Taken together, the data from the ANOVA and PCA analyses support the view that activities in facial muscles indicate infants can discriminate pleasant from unpleasant tastants.

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Limitations of the study

Besides providing substantial evidence regarding which muscle regions exhibited increased activity in response to the 3 tastants and the likely facial expressions that occurred, the results indicated that the high frequencies of response to quinine (96.7%), citric acid (96.7%), and sucrose (80%) supported the proposal that multi-electrode EMG has the potential to become a method for assessing taste function in infants (Table 4). Key factors needing further research in order to provide an acceptable method include 1) optimizing the positions of the electrodes as reported by Tassinary et al. (1989) and Horio (2003) so that a grid or map of an infant's face can be used to position the electrodes, 2) establish the reliability of the data through replication of stimuli within an assessment session, 3) compare the activity, muscles identified and proposed facial actions using electrodes with videotaped FACS data collected at the same time in the same infants to further verify the data, and finally 4) optimize tastant concentrations and interstimulus and rinsing times. These are challenging tasks illustrated by the fact that, to date no reliability data is available for studies of taste-elicited facial expressions in infants including those using the FACS coding system, and no data exists that shows the appropriate concentrations of the common tastants to use with infants.

Conclusion

It is concluded that most of the 5–8-week-old infants assessed could perceive and discriminate between the pleasant taste of sucrose and the unpleasant tastes of citric acid and quinine and there was evidence that indicated that they can discriminate between the 2 unpleasant tastants. The study confirmed that infants use a variety of facial muscle regions when responding to taste stimuli and that the multi-electrode EMG system may be an appropriate technique for capturing the arrays of muscle activities that is very difficult to observe by eye as required using video-imaging procedures. The high percentage of infants responding to each of the 3 tastants and the substantial amount of information on facial expressions and their hedonic meaning suggests that EMG may be suitable for use in tests of taste sensitivity, discrimination and hedonics of taste stimuli that would contribute to knowledge of taste development in young humans. It may also be suitable for use in a clinical test of taste for infants in which the measurement of the activity of multiple facial muscle regions simultaneously would be undertaken to assess taste function and minimize the chance of an individual being mistakenly classified as a "non-responder".

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